

Alif Jawad

Tucson, Arizona, United States

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RESEARCH PROFILE

Ph.D. student in Mechanical Engineering with research experience in Flash Joule Heating synthesis of MAX phases, MXene-related materials, electrochemical materials, materials characterization, molecular dynamics simulations, and machine-learning-assisted materials modeling. Current research focuses on rapid synthesis pathways for low-entropy and high-entropy MAX phases, molten-salt-assisted processing, and structure-property relationships in advanced functional materials.

EDUCATION

University of Arizona

Ph.D. Student in Mechanical Engineering

2024 – Present

Tucson, Arizona, United States

- **Advisor:** Dr. Majid Beidaghi
- **Current GPA:** 3.75/4.00
- **Research Focus:** Flash Joule Heating synthesis of MAX phases, MXene-related materials, electrochemical materials, and materials characterization.

Bangladesh University of Engineering and Technology

B.Sc. in Mechanical Engineering

2018 – 2023

Dhaka, Bangladesh

- **GPA:** 3.53/4.00
- **Undergraduate Thesis:** Analysis of tensile strength of high-entropy alloy FeCrCuCoNi using molecular dynamics simulations and machine learning.

RESEARCH INTERESTS

- MXenes and MAX phases
- Flash Joule Heating synthesis
- High-entropy ceramics
- 2D materials
- Electrochemical energy storage
- Materials characterization
- Molecular dynamics simulations
- Machine learning for materials science

RESEARCH EXPERIENCE

Ph.D. Researcher, Beidaghi Electrochemistry Research Group

Supervisor: Dr. Majid Beidaghi

2024 – Present

University of Arizona

- Developing and optimizing Flash Joule Heating synthesis routes for Ti-, Nb-, and V-based MAX phases, with emphasis on precursor composition, reaction temperature profile, dwell condition, and phase purity.
- Investigating synthesis pathways from low-entropy to high-entropy MAX phases using rapid thermal processing and molten-salt-assisted reaction environments.
- Performing structural and chemical characterization of synthesized materials using X-ray diffraction, scanning electron microscopy, energy-dispersive spectroscopy, Raman spectroscopy, and related techniques.
- Analyzing phase evolution, impurity formation, and microstructural differences between Flash Joule Heating and conventional furnace synthesis routes.
- Collaborating on electrochemical evaluation of silicon–MXene ink electrodes for energy-storage applications.
- Collaborating on the preparation of delaminated MXene films with different intercalants for membrane and filtration-related applications.

- Conducted molecular dynamics simulations to calculate tensile strength of FeCrCuCoNi high-entropy alloys under varying compositional conditions.
- Generated low-fidelity single-crystal and high-fidelity polycrystalline simulation datasets for machine-learning-assisted mechanical property prediction.
- Trained a multi-fidelity physics-informed neural network model to predict tensile strength from composition-dependent molecular dynamics data.
- Analyzed simulation errors and fidelity relationships to improve predictive performance for high-entropy alloy strength estimation.

PUBLICATIONS

Journal Article

2025

N. E. Chowdhury, A. **Jawad**, A. Rahman, and M. J. A. Khan

- “Multi-fidelity neural network–based prediction of tensile strength of high-entropy alloy FeNiCoCrCu using molecular dynamics data.” *Journal of Molecular Modeling*, 31(8), 1–9.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Fundamentals of Materials for Engineers

Fall 2024 – Spring 2026

AME/MSE 331R, Course Instructor: Dr. Majid Beidaghi

University of Arizona

- Graded homework, course projects, and final assessments for multiple offerings of an undergraduate materials science course.
- Led problem-solving sessions and provided office-hour support for students on topics including crystal structures, phase diagrams, mechanical properties, and materials processing.
- Supported students in completing final course projects by providing technical guidance, feedback, and instructional resources.
- Prepared recorded instructional materials to assist students with project expectations and course problem-solving strategies.

AWARDS, GRANTS, AND PROFESSIONAL DEVELOPMENT

NSF-Funded Workshop Travel Scholarship

May 2026

Unravelling Atomistic Correlations in Quantum Materials Workshop

Arizona State University, Tempe, AZ

- Awarded a merit-based \$750 travel scholarship to attend a two-day NSF-funded workshop on quantum materials, electron microscopy, X-ray-based characterization, ultrafast diffraction, cryogenic TEM, and quantum metrology.

RISE Undergraduate Research Grant

2022

Research and Innovation Centre for Science and Engineering, BUET

Dhaka, Bangladesh

- Awarded a research grant of BDT 74,880, approximately USD 700, to support undergraduate thesis research on mechanical property prediction of alloy materials using molecular dynamics simulations and machine learning.

TECHNICAL SKILLS

Materials Characterization: XRD, SEM, AFM, EDS, Raman Spectroscopy, UV-Vis Spectroscopy, TGA

Electrochemical Characterization: Cyclic Voltammetry, Galvanostatic Charge-Discharge, Electrochemical Impedance Spectroscopy

Synthesis and Processing: Flash Joule Heating, solid-state synthesis, molten salt processing, glovebox operation, fume hood operation, centrifugation, FlackTek SpeedMixer

Simulation and Modeling: LAMMPS, MATLAB, ANSYS Workbench, molecular dynamics simulations, machine learning, deep learning

Programming: Python, C, Arduino

Data Analysis and Visualization: Origin, OVITO, Tecplot 360, Excel, PowerPoint, LaTeX

CAD and Design: SolidWorks, AutoCAD

PROFESSIONAL EXPERIENCE

Industrial Attachment Trainee

Unilever Bangladesh Limited

2022

Chittagong, Bangladesh

- Calculated cooling load requirements for a production floor and incorporated measured dimensions and calculations into existing technical drafts.
- Tabulated process water-loss data and supported optimization efforts for water-use reduction in industrial operations.

LEADERSHIP AND SERVICE

Public Relations Officer

Bangladesh Students Association

2025 – 2026

University of Arizona

- Organized student association events and coordinated activities for the Bangladeshi student community at the University of Arizona.
- Managed outreach and communication activities to increase engagement and participation in association programs.

Team Captain

Hall Volleyball, Badminton, Basketball, and Table-Tennis Teams

2022

BUET

- Organized and managed multiple hall sports teams, leading several teams to semi-final placements in university-level tournaments.
- Coordinated team logistics and created hall funds to support player development and team activities.

Top 140 National Participant, Ennovators 6.0

Business Case Competition by Banglalink

2022

Bangladesh

- Selected among the top 140 national participants after several rounds of a competitive business case competition.

REFERENCES

References available upon request.